

Market Access and Compliance innovation for AI-Based Functional Safety Systems

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Abstract—The IEC Test Report Format (TRF) standardizes product testing and compliance, ensuring consistency across international standards and simplifying global certification. However, the increasing complexity of Artificial Intelligence (AI) and Functional Safety (FuSa) and Cybersecurity demands a more advanced and comprehensive approach. We introduce the Compliance Report Format (CRF)—an enhanced framework that builds upon IEC TRF by integrating AI assurance and FuSa compliance management. The CRF improves traceability, consistency, and efficiency in assurance case development and certification. To achieve this, the CRF incorporates key enhancements, including a profile-based approach, support for agile methodologies, explicit linkages to evidence, and integration with both the safety plan and assurance case. It also introduces structured arguments, synergy aspects, practical guidelines, and references to relevant research. With safety cases, a subset of assurance case, becoming increasingly central in UL 4600 “automotive autonomy” and the forthcoming IEC 61508 “generic Functional Safety (FuSa)” third edition, the CRF aligns with evolving regulatory landscapes. It also supports IOS/IEC TR 5469 “AI and FuSa”, strengthening the integration of AI and safety. By promoting a shift-left approach, the CRF facilitates early AI risk analysis, FuSa integration, and continuous validation, reducing late-stage compliance burdens. Additionally, by leveraging agile practices and the definition of ready concept, it minimizes compliance overhead while enhancing traceability of AI-related safety evidence. CRF’s profile-based methodology tailors AI and FuSa requirements to domain-specific standards, ensuring flexibility and adaptability. Development engineers benefit from structured guidelines, references to industry standards, and links to scientific research, streamlining compliance efforts. By modernizing the compliance landscape, the CRF empowers engineers, auditors, and certification bodies to navigate AI-driven safety and cybersecurity challenges more effectively.

Keywords—AI safety system, worldwide compliance, safety case, market access

I. INTRODUCTION

In recent years, the growing adoption of Artificial Intelligence (AI) within safety-critical systems has introduced new challenges in meeting regulations alongside AI and FuSa standards. Traditionally, the IEC TRF has been an essential tool for documenting compliance with standards. However, the increasing complexity of modern systems—driven by AI integration—necessitates a more structured, evidence-driven, and adaptable approach to compliance reporting.

To address these challenges, a profile-based, agile, and systematically integrated Compliance Report Format (CRF) that facilitates cross-standard alignment and verification of AI and FuSa requirements. The CRF leverages agile methodologies, structured argumentation, cross-standard

synergies, and strong links to safety plans and evidence to reduce redundancy, documentation overhead, and certification costs while ensuring compliance with standards such as ISO/IEC TR 5469:2024 “*Artificial intelligence - Functional safety and AI systems*” and IEC 61508:2010 “*Functional Safety of Electrical/Electronic/Programmable Electronic Safety-related Systems (E/E/PE, or E/E/PES)*”. This raises the first research question:

- RQ1: How can the development of a CRF reduce the burden of complying with AI, cybersecurity and Functional safety standards?

Furthermore, effective communication between manufacturers and certification bodies is essential for a smoother certification process. The CRF seeks to enhance clarity, transparency, and collaboration, leading to improved regulatory interactions. Thus, the second research question is:

- RQ2: What specific features in the CRF can enhance clarity and improve communication both internally and with certification bodies, ensuring a smoother certification process?

Beyond compliance and communication, maintaining a clear record of the reasoning behind design and development decisions is vital for long-term knowledge retention. The CRF will implement a documentation strategy that captures design rationale and decision-making processes, mitigating knowledge loss due to workforce turnover or retirement. This leads to the following research inquiry:

- RQ3: How can a structured CRF effectively document and trace the rationale behind design and coding decisions to mitigate knowledge loss?

Developing a comprehensive safety case or security case is a resource-intensive process that often results in significant workload and inefficiencies. The Compliance Report Format (CRF) addresses these challenges by incorporating agile practices, reusable profiles, and structured linkages to safety plans and supporting evidence. This integration reduces redundancy and documentation overhead, while ensuring consistency and traceability across compliance activities. Moreover, traditional regulatory processes frequently involve duplicative documentation efforts, particularly when aligning with multiple standards. The CRF aims to identify and apply streamlined documentation methodologies that preserve full traceability and meet the rigorous requirements of AI, Cybersecurity and functional safety standards. This leads to the following research question:

- RQ4: How can the use of profiles and agile practices in the CRF reduce the workload required to provide sufficient evidence while ensuring compliance with AI, cybersecurity and safety standards?

Lastly, as AI and FuSa regulations evolve, organizations must navigate multiple overlapping standards. By leveraging synergies across FuSa, cybersecurity, and AI regulatory frameworks, the CRF aims to integrate and align requirements across different regulatory landscapes, posing the final research question:

- RQ5: How can the CRF leverage cross-domain knowledge to integrate and align requirements across different standards, such as IEC 61508 and ISO 26262, while identifying synergies and managing potential conflicts?

By addressing these challenges, the proposed CRF emphasizes structured profiles, agile methodologies, evidence-based compliance, and cross-domain synergies, facilitating the adaptation of compliance requirements to the unique contexts of AI-based systems. It supports safety engineers, auditors, and other stakeholders in creating structured, evidence-backed safety cases that meet the evolving demands of global regulatory frameworks. This paper explores the historical evolution of compliance formats, identifies existing gaps, and proposes a forward-looking approach to managing the complexity of AI and functional safety systems.

This paper first introduces foundational concepts, including relevant definitions and the IEC TRF approach, to establish a common understanding of compliance frameworks in AI-based functional safety systems. As part of this work, we have developed new and refined definitions—such as those for the CRF and profiles—based on systematic assessments of safety standards and collaborative research activities. These definitions are tailored to the unique challenges of AI integration and support a more structured and auditable compliance process. Additionally, this paper outlines key AI, cybersecurity and safety standards that form the regulatory basis for our work. In the subsequent sections, we present the research methodology used to address the identified research questions, combining a thorough review of standards and literature with empirical insights from industry collaboration.

This paper introduces the Compliance Report Format (CRF), designed to address these challenges through several innovative enhancements:

- **Profile Approach:** Tailors compliance requirements to specific domains, roles, and contexts, improving clarity and applicability.
- **Adapted Agile and adapted agile practices:** Integrates agile methodologies to streamline compliance activities and improve responsiveness.
- **Link to Evidence, Work Products, Safety Plan, and Safety Case Chapters:** Establishes explicit connections between evidence, relevant documentation, safety plans, and specific safety case chapters, enhancing traceability.
- **Arguments:** Employs structured argumentation to clearly articulate compliance rationales and facilitate robust safety justification.
- **Functional Safety, Cybersecurity and AI Requirements, Synergies, and Similarities to Other Standards:** Exploits overlaps and identifies synergies across functional safety, cybersecurity, and AI

standards, reducing redundancy and enhancing cross-standard compliance.

- **Guidelines and References:** Provides comprehensive guidelines and accessible references to support consistent understanding and streamlined application of requirements.

II. BACKGROUND

A. Definitions

Below, we list key definitions essential for a common understanding in compliance and functional safety, especially in AI-based systems. Standardized terminology improves communication between engineers, regulators, and certification bodies, reducing ambiguity and misinterpretations. Clear definitions prevent regulatory inconsistencies, avoiding certification delays and compliance burdens while enhancing transparency and interoperability across standards.

What is a TRF? According to [1] “*A TRF documents the actual results from tests performed on a product or component to assess conformity with a standard. It is prepared by a National Certification Body (NCB). TRFs developed by the IEC/IEEE are complete and perfectly aligned with IEC Standards. Further, we know that they have been fully verified and endorsed by recognized Test Laboratories and Certification Bodies before public release. In other words, they are fully validated.*”

As part of this project we have developed this definition for the CRF: A Compliance Report Format is a structured and auditable framework designed to document and demonstrate adherence to specific standards, such as cybersecurity, functional safety, and regulatory requirements. CRF acts as a streamlined compliance safety case for one standard that integrates evidence, argumentation, and relevant documentation to validate compliance efficiently (see figure 1 below). While an ordinary, e.g. system safety case includes CRFs for several standards. A CRF often includes many references. E.g. the work products in ISO 26262 standards “*Road vehicles – Functional safety*”. ISO 26262-6 “SW” includes 25 work products, some of which are refined versions. In practice, ‘refined versions’ typically refer to new editions of existing work products (documents) based on earlier versions from either this part of the standard or other parts in the standard series, such as ISO 26262-4 “*product development at the system level*”. This means that a future CRF for the ISO 26262-6 may include reference to these work products.

Standards like ISO 26262 series, ISO 21448:2022 “*Road vehicles — Safety of the intended functionality*” and ISO/PAS 8800:2024 “*Road vehicles — Safety and artificial intelligence*” lists many work products (WP). This helps in the planning of which documents that should be developed WP are references in a safety case. ISO 26262-1:2018 Automotive 3.185 work product: “*documentation resulting from one or more associated requirements of ISO 26262. Note 1 to entry: The documentation can be in the form of a single document containing the complete information for the work product or a set of documents that together contain the complete information for the work product*”. Some of the most relevant WPs are: Safety plan, Description of the System (DoS), Operational Design Domain (ODD) and Concept of Operation

(ConOps) and the Hazard log. For further information see [6] and [9].

Slightly different definitions exist for safety cases in the different safety standards. From a pragmatic approach, the definition in ISO 26262 is chosen as it includes work products as part of the definition. ISO 26262-1:2018 3.136 safety case: “*argument that functional safety is achieved for items, or elements, and satisfied by evidence compiled from work products (documents/information) of activities during development.*” The CRF’s will normally be referenced in the safety case. There will normally be one CRF for each part of the IEC 61508 requirements parts (Part 1, 2 and 3).

A profile is a selected set of characteristics from a standard, tailored for specific applications. Profiles should create when necessary for technical, regional, or application needs. Examples include custom adaptations of standards for specific industries, such as fire alarm systems. User-specific profiles may apply to manufacturers and operators. Some profiles are designed for a single function. Others define security requirements, such as meeting IEC 61508 at SIL2 (Safety Integrity Level).

B. Functional safety, Cybersecurity and AI standards

Below we have described three of the most relevant standards for our assessment and research projects.

IEC 61508:2010 mandates compliance with safety requirements, ensuring each clause’s objectives are met. If no product or application standard exists, exemptions may be justified. The standard outlines FuSa requirements for E/E/PE safety systems. It defines a structured safety lifecycle covering design, implementation, maintenance, and decommissioning across industries. Its core framework includes hazard analysis, safety requirements allocation, validation, and maintenance. SIL quantifies safety integrity, while a risk-based approach aligns safety measures with risks. The standard emphasizes documentation, verification, and systematic safety measures to prevent faults and support sector-specific standards. Annex A of IEC 61508-1 “*General requirements*” provides a structured documentation approach under Clause 5. It categorizes documents by lifecycle phases, such as hazard analysis, safety requirements, validation, and system integration, ensuring clear, complete, and accessible documentation for functional safety compliance.

ISO/IEC TR 5469:2024 explores the integration of AI into safety-critical systems, emphasizing its implications for functional safety. The report identifies challenges like transparency, environmental complexities, and risks from adversarial inputs. It introduces the three-stage realization principle for AI systems—data acquisition, knowledge induction, and output generation—and discusses the limitations of current safety standards in addressing AI-driven systems. The document outlines properties, risk factors, and methods for AI verification, including the classification of AI technologies based on usage and compliance. It offers strategies for mitigating AI-specific risks, such as data drift, adversarial attacks, and system explainability. The report aims to guide developers in applying AI technologies to ensure safety, detailing control measures, functional safety artifacts, and methodologies to align AI systems with existing international standards like IEC 61508.

Underwriters Laboratories (UL) has issued already the third edition of UL 4600:2023 “*Standard for Evaluation of*

Autonomous Products”. This safety standard focusing on autonomous product evaluation, emphasizing comprehensive safety case development and assessment. It outlines guidelines for risk assessment, fault management, interaction with humans, and lifecycle considerations, alongside ensuring compatibility with other standards like ISO 26262 and ISO 21448. The last edition introduces revised requirements for safety cases, fault analysis, sensors, object tracking, and adds provisions for failsafe mechanisms. It stresses flexibility, allowing varied methods for compliance without mandating specific design processes. The standard offers extensive prompts for assessing fully autonomous systems operating without human intervention, aiming to ensure structured, consistent safety practices across diverse operational contexts.

The ISA/IEC 62443 series of standards, developed by the ISA99 committee since 2002, address cybersecurity for industrial automation and control systems (IACS). These standards, adopted globally by the IEC and endorsed by the UN, apply across industries, ensuring security for operational technology. Recognized as a horizontal standard in 2021, ISA/IEC 62443 defines best practices, risk assessment processes, and certification mechanisms. It emphasizes shared responsibility among asset owners, suppliers, integrators, and service providers. By bridging operations and IT, the standards provide a holistic approach to securing control systems throughout their lifecycle in various sectors.

III. METHODS

To answer the five research questions, we have followed a research process by systematically studying relevant safety standards, safety regulations, the EU AI Act (Regulation 2024/1689), and recent research literature on the topic of AI and functional safety use in different domains (A). In addition we have included information from collaboration with companies and lessons learned (B).

A. Review of standards, regulations, and literature

The relevant safety and AI standards were identified by ongoing participation in IEC 61508 and ISO/IEC TR 5469 committee work, discussions with colleagues, and searches for relevant ISO, IEC, SAE, and UL standards. The safety standards evaluated are IEC 61508:2010, IEC 61508 draft 2025, UL4600:2023, ISO 26262:2018, ISO 21448:2022, ISO/IEC TR 5469:2024 and ISO/PAS 8800. Regulations and AI Act has been reviewed as part of former and ongoing AI, railway, seaborne and automotive research projects. To identify relevant research literature, we performed a quick literature review, searching through IEEE Xplore, Google Scholar, and ChatGPT 4.5 with combinations of the terms “safety case” and “compliance.” Most of the research papers focuses on safety cases and lacks information on TRF and work products. Myklebust et al. [2], Koopman [3], and EN 50129 :2018/AC:2019-04 “*Railway applications - Communication, signalling and processing systems - Safety related electronic systems for signalling*” includes relevant chapters and topics but no information regarding the safety case references. Only [2] include AI chapters, but Koopman [3] covers autonomous systems.

B. Industry Collaboration and Lessons Learned

This paper employs a qualitative, practice-oriented research methodology involving active collaboration with multiple industry partners through a series of structured projects and discussions. Throughout these collaborative engagements, we systematically documented practical

insights as "lessons learned," capturing valuable empirical observations, critical challenges faced, and effective solutions implemented. These insights were subsequently analyzed, synthesized, and articulated clearly as actionable solutions detailed in Chapter IV.

IV. RESULTS

The evolving landscape of safety, cybersecurity, and AI standards demands structured approaches to ensure reliability, security, and regulatory compliance. As industries integrate automation, AI, and functional safety systems, adapting to international standards is essential. This paper examines the state of IEC Technical Report Forms (TRFs) and the need for a profile-based approach to tailor generic standards across sectors, roles, and applications. It highlights accountability in distributed development and explores challenges in aligning agile methodologies with safety-critical requirements. This paper also underscores the growing role of assurance cases in AI validation, reflecting global regulatory trends. By linking arguments, work products, and safety cases, it provides a framework for maintaining traceability and compliance.

Finally, it addresses synergies and conflicts between safety and security standards and offers guidance for organizations adopting functional safety and AI management frameworks. This structured approach helps industries navigate safety, cybersecurity, and AI integration in an increasingly digital world.

A. Current editions of IEC TRF

IECEE, the IEC System for Conformity Assessment Schemes for Electrotechnical Equipment and Components, provides internationally recognized certification based on IEC standards, facilitating mutual recognition of test results among member countries. By reducing the need for repeated testing, the IECEE schemes help manufacturers accelerate market entry, cut costs, and overcome international trade barriers. The IECEE CB (Certification Bodies) scheme particularly allows manufacturers to use a single test and certificate to gain global market acceptance

As of now, no official IEC TRF exists for functional safety or AI standards. However, several TRFs are available for relevant parts of the IEC 62443 series on cybersecurity. We developed our own TRF for the IEC 61508 series in 2005 and have been using them in certification projects ever since. After one year we included a link to ISO 9001 "quality management" as this was great help for startups and Small and Medium Enterprises (SMEs). We received an accreditation for our approach in 2006. In 2007, we included links to guidelines like [4] and ISO 90003 "Software engineering — Guidelines for the application of ISO 9001:2015 to computer software".

The current IEC TRF for IEC 62443 includes these aspects: **conformity statement** (brief description how the requirements is met), **conformity evidence** (documentation evidence to support the conformity statement), **Result – Remarks** (when N/A (Not Applicable) is used, a comment describing why N/A is used) and **Verdict** (Vales: N/A, P (Pass), F (FAIL)).

In the Table below, we have listed the current (2025-03-20) TRFs issued by IEC. They consist of an introduction part in word and a excel sheet with the requirements.

TABLE I. IECEE TRF DOCUMENTS

TRF	YEAR	EDITION
IECEE TRF 62443-2-4C. THIS TRF APPLIES TO: IEC 62443-2-4:2015, IEC 62443-2-4:2015/AMD1:2017 PART 2-4: SECURITY PROGRAM REQUIREMENTS FOR IACS SERVICE PROVIDERS	2023	4
IECEE TRF 62443-2-4A. THIS TRF APPLIES TO: IEC 62443-2-4:2015 PART 2-4: SECURITY PROGRAM REQUIREMENTS FOR IACS SERVICE PROVIDERS	2017	1
IECEE TRF 62443-3-3B. THIS TRF APPLIES TO: IEC 62443-3-3:2013 SECURITY FOR INDUSTRIAL AUTOMATION AND CONTROL SYSTEMS – PART 3-3: SYSTEM SECURITY REQUIREMENTS AND SECURITY LEVELS	2023	2
IECEE TRF 62443-4-1B. THIS TRF APPLIES TO: IEC 62443-4-1:2018 PART 4-1: SECURE PRODUCT DEVELOPMENT LIFECYCLE REQUIREMENTS	2023	2
IECEE TRF 62443-4-2B. THIS TRF APPLIES TO: IEC 62443-4-2:2019 PART 4-2: TECHNICAL SECURITY REQUIREMENTS FOR IACS COMPONENTS	2023	2

B. Profile approach

Generic standards like IEC 61508 and ISO/IEC TR 5469 can often be adapted to sectors, applications, roles (e.g. OEM (Original Equipment Manufacturer), Tier1 or operator and internal roles [5]) and environments. This approach is in line with the standard for cybersecurity IEC TR 62443-1-5:2023 "Security for industrial automation and control systems - Part 1-5: Scheme for IEC 62443 security profiles". This is also an important part of distributed development as it is of crucial importance who have responsibility for the actual requirements. Only the ISO 26262 series includes relevant information and requirements related to distributed development. Distributed development is defined in ISO 26262-1:2018 as "development of an item or element with development responsibility divided between the customer and supplier(s) for the entire item or element". Requirements regarding distributed development are presented in part 6 "Software development" and part 8 "Supporting processes". The contract, named "Development Interface Agreement" (DIA) is then of great importance and described in Annex B of ISO 26262-8. The DIA has been improved for AI safety systems in [6]. Other approaches is often a contract between the manufacturer/supplier and the operator for large projects as e.g. European Railway Traffic Management System (ERTMS) for railway.

When applying this approach, one should understand the possible pitfalls. Generic pitfalls could be "Responsibility Gaps in Distributed Development:" and "Contractual and Interface Complexities". The safety pitfalls could be "Failures in Safety Responsibility Delegation" and "Cybersecurity Vulnerabilities Undermining Safety". While the AI pitfalls could be "Integrating AI Safety with Evolving Regulations", "Explainability and Transparency challenges" and "machine learning can have non-intuitive decision clusters".

This part of the CRF reduces the complexity of applying generic standards to diverse roles, domains, and architectures—enabling clearer responsibility allocation and more manageable compliance activities across distributed development environments. New column titles in a n excel

sheet or headings in a word documents are, see also Table II below: Roles and Domain/topics.

C. Software, AI, Cybersecurity and safety practices

As part of this work we have mainly adapted and included agile practices [7] and [8] in our assessment and research. Some of the challenges we have encountered is definition of ready and definition of done [9] as the system shall not be taken into operation before the Certificate has been issued or the safety case has been accepted. In addition, we have implemented FuSa sprints [10] to ensure that the design fulfils requirements and that the work products are finished in due time. Relevant pitfalls related to practices are “Incompatibility Between Agile and Safety Standards”, “DevOps (developers and operations teams collaborate closely, using automation and continuous feedback to build, test, and deliver software faster) and Agile for Safety-Critical Systems” and “Balancing Agility with Traceability and Risk Management”. AI practices relevant to software developers include AI-assisted code generation for reducing errors, AI-driven test automation for efficient verification. Explainable AI (XAI) ensures transparency in decision-making. AI-driven backlog prioritization optimizes sprint planning. AI also automates documentation generation, supporting compliance in regulated domains. These practices integrate with Agile and DevOps while maintaining safety and traceability requirements. Wang et al. [11] conducted an industrial case study on communication channels in safety analysis, highlighting their significance in safety-critical system development. The study, based on surveys and interviews with industry experts, identified key communication methods such as formal meetings, documentation, emails, and project coordination tools. It also outlined challenges, including fragmented and inconsistent safety information, security concerns, and miscommunication between developers and safety analysts. The findings emphasize the need for structured communication strategies to enhance safety analysis effectiveness. This shows that applying agile communication practices like daily scrum [7] are important.

The **shift-left approach** integrates verification, validation, and compliance **early** in development, preventing costly rework and late-stage failures. In our CRF approach we also integrate arguments, links to safety plan and safety case and agile practices adapted to safety, cybersecurity and AI. This is crucial for AI-based functional safety systems, where early evidence ensures safety and regulatory compliance. Closely linked to Definition of Ready (DoR) and Definition of Done (DoD), shift-left ensures well-defined safety and security requirements before development (DoR) [9] and completed validation, including safety cases and compliance checks, before release (DoD). In regulated domains, shift-left aligns with Continuous Assurance and Agile Compliance, embedding certification and risk assessment into iterative cycles. This CRF approach structures **early evidence** collection, enhancing traceability, stakeholder alignment, and safety argumentation. This method also integrates with DevOps for regulated domains, automating AI, cybersecurity and safety checks within **CI/CD (Continuous Integration / Continuous Deployment)** pipelines.

New column titles in an excel sheet or headings in a word documents are, see also Table II below: Practices.

D. Agile Communication Practices

Co Effective communication practices have consistently proven essential across all our research and assessment initiatives. In addition to implementing Definition of Ready (DoR) and Definition of Done (DoD) as previously outlined, the following agile communication practices should be considered and integrated into relevant projects to ensure clarity, efficiency, and successful collaboration:

- **Daily Scrum:** Short daily meetings to discuss progress, plans, and obstacles, ensuring continuous communication within the team [9].
- **Sprint Planning:** Regular sessions where teams collaboratively plan the tasks and priorities for the upcoming sprint.
- **Sprint Review:** Meetings held after each sprint to review completed work and gather feedback directly from stakeholders.
- **Sprint Retrospective:** Dedicated reflection meetings after each sprint to identify improvements and communication enhancements.
- **Backlog Refinement:** A continuous practice involving regular review, clarification, and prioritization of backlog items to ensure clear and effective communication of requirements. Our experience indicates that the interpretation and understanding of requirements, such as cybersecurity requirements outlined in IEC 61508 (see e.g. clause 7.4.2.3 in IEC 61508-1), evolve considerably over the course of long-term development projects spanning several years.
- **Task Board:** Visual management tool displaying tasks, status, and team responsibilities, enhancing transparency and immediate communication.
- **Information Radiators:** Public displays of critical project information within the team's workspace to enhance real-time communication and awareness.
- **Hazard Stories:** Specialized user stories explicitly designed to communicate potential hazards, associated risks, and required safety precautions. These stories not only identify critical safety scenarios but also facilitate early identification of preventive measures and system barriers, enhancing safety awareness and proactive risk management [12].
- **Safety Stories:** User stories explicitly designed to communicate clear and detailed safety requirements to all stakeholders, helping to ensure that safety considerations are integrated early and continuously throughout the development process [13].
- **Rapid Prototyping:** Quickly developed prototypes communicated to stakeholders for rapid feedback, refining requirements effectively. Our latest research projects have shown that this approach has become more popular, and we include often a W approach [6].

E. Linking Evidence, CRF, Work Products, and the Safety Case

When we initiated the CRF approach in 2005, it primarily served as the foundation for certification reports, i.e. the CRFs were referenced in the certification reports.

In recent years, the importance of assurance cases has grown significantly. In 2020, UL in the USA adopted safety cases as the core approach for UL 4600, the Standard for Safety for the Evaluation of Autonomous Products, emphasizing their critical role in AI system assurance. Similarly, in 2022, a German DIN/DKE “*Deutsches Institut für Normung/ German Commission for Electrical, Electronic & Information Technologies of DIN and VDE*” document [14] recommended the assurance approach, recognizing its necessity for AI systems that often lack interpretability and standardized validation, making reliability difficult to ensure. Assurance cases, proven effective in other technological domains, enhance AI safety by leveraging field data to refine arguments and identify risks early. However, as AI continues to evolve, further research is required to improve their structure and validation methods to meet stringent safety standards. The maritime industry has traditionally not used the assurance approach, but in 2023, DNV issued a recommended practice [15] advocating for assurance cases. Assurance cases encompass various aspects, including safety cases and cybersecurity cases. In [2], key topics essential for inclusion in a safety case are outlined. The UK has a long-standing history of utilizing safety cases as part of a goal-based regulatory framework. This experience underscores the growing relevance of assurance cases in AI and other emerging technologies, where structured, evidence-based safety justifications are becoming increasingly necessary [16].

Ensuring traceability from the safety plan to the safety case—and vice versa—is crucial for maintaining consistency, completeness, and compliance. Bidirectional traceability guarantees a seamless transition from planning to assurance, which is particularly vital for AI-driven safety systems where ongoing compliance and updates are required. A list of relevant work products should be included in the Safety Plan or in supporting documents such as the documentation plan.

New column titles in an excel sheet or headings in a word documents are, see also Table II below: Evidence and links.

The figure below illustrates the process linking work products, the CRF, and a safety case.

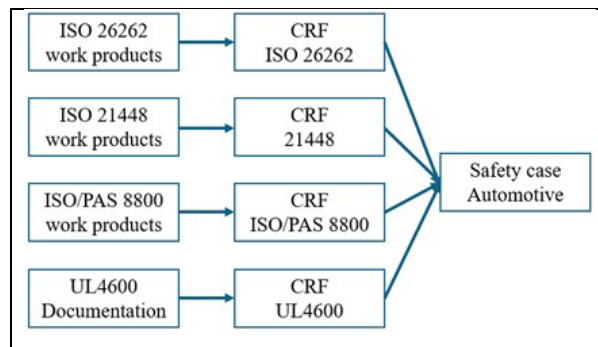


Fig. 1. Example of the process from work products, CRF to a safety case

Table II below illustrates the relationships among the different parts of the safety case, associated work products, and corresponding CRFs, clearly demonstrating their integration and traceability.

TABLE II. SAFETY CASE, CRF AND WPS

Safety case parts	CRF references and WP comments
0: Introduction	The WP Safety plan is referenced
1: Definition of System (DoS)	The WP DoS is referenced.
2: Operational Design Domain (ODD)	The WP ODD is referenced.
3: Concept of Operation (ConOps)	The WP ConOps is referenced.
4: Quality Management Report (QMR)	Several to many WPs are referenced. Reference to test&Verification&Validation (TVV) and analysis reports, specifications, reviews and audits. Arguments included when relevant.
5: Safety Management Report (SMR)	Several to many WP are referenced. Reference to TVV and analysis reports, specifications, reviews and audits. Arguments included when relevant.
6: Technical Safety Report (TSR)	Several to many WP are referenced. Reference to TVV and analysis reports, specifications, reviews and audits. Arguments included when relevant.
7: AI Safety Report (ASR)	Several to many WPs are referenced. ISO/PAS 8800:2024 includes altogether 28 WPS including refined versions. Reference to TVV and analysis reports, specifications, reviews and audits. Arguments included when relevant.
8: Human aspects and oversight (HAO)	Expect several WP to be mentioned in future standards. Reference to TVV and analysis reports, specifications, reviews and audits. Arguments included when relevant.
9: Related Safety Cases (RSC)	Safety case for products and systems are referenced. Certificates are referenced. Certificate reports are referenced
10: Conclusion	Summary and conclusions provided in the CRF, along with relevant references. The CRF explicitly lists all its internal references, facilitating impact analysis and aiding both manufacturers and certification bodies in clearly identifying necessary actions.

The figure below illustrates the iterative linkage process. Initially, the Safety Plan defines the scope and the required Work Products (WPs). The first draft of the DoS should also be issued early, as experience shows that engineers often lack sufficient initial knowledge about complex safety systems to be developed. Subsequent versions of the Safety Plan and WPs are developed iteratively. These WPs feed into the Compliance Report Format (CRF), which is developed concurrently with Evidence & Documentation, ensuring traceability. The final Safety Case integrates all documentation, ensuring robust validation. To maximize efficiency, the chapters of the Safety Plan should preferably

align structurally with the parts of the Safety Case. As recommended in reference [6], adopting an agile Safety Plan approach and maintaining early visibility of the Safety Case structure enhances efficiency and promotes synergy.

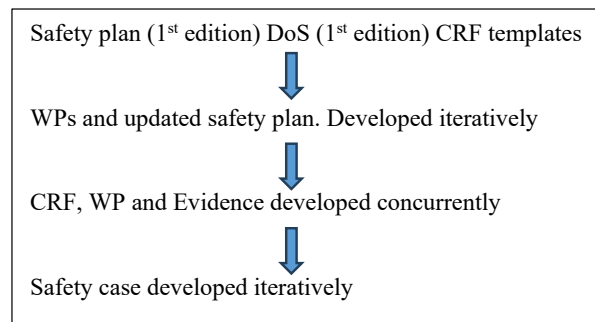


Fig. 2. Illustrative linkage between safety plan, WP, CRF, evidence and safety case

F. Arguments

An argument is a structured claim supported by evidence; without argumentation, evidence remains unexplained and therefore insufficient. Merely presenting facts or data does not adequately justify claims about product and system safety. Manufacturers must clearly articulate why their products and systems meet safety standards. This requirement directly addresses Research Question 3 (RQ3), emphasizing the need to document and trace the rationale behind design and coding decisions effectively. Both our literature review and discussions with software engineers underscore the importance of maintaining robust arguments, also due to the occurrence of knowledge loss when engineers retire or transition to other roles. Structured arguments provide several benefits: they enhance developers' understanding of safety considerations, improve analytical depth, facilitate clear communication with certification bodies, and establish a rigorous foundation for assurance cases. The SCSC Guide [17] particularly recommends employing modular assurance and dialectic arguments—where both supporting and opposing viewpoints are analysed—to improve clarity and mitigate bias. Additionally, ISO/PAS 8800:2024 Annex B introduces a specialized assurance argument structure specifically designed for AI systems.

Consider a bad example from a recent assessment: "The measurement value was outside the specified tolerance. However, due to measurement uncertainty, it is reasonable to expect that the true value falls within acceptable limits" (see also the figure 2 below). Utilizing defeaters in arguments is crucial, as they ensure arguments remain strong, credible, and logically robust. A defeater serves as a critical piece of evidence or reasoning that challenges or weakens a claim. Incorporating defeaters helps reveal potential gaps or flaws in reasoning, thereby ensuring only well-supported conclusions withstand scrutiny.

Key pitfalls in the application or misuse of arguments within the Compliance Report Format (CRF) and safety cases include insufficient understanding of risk, over-reliance on outdated or incomplete data, inadequate communication among stakeholders, unclear or overly extensive documentation [18], insufficient analysis of change impacts [19], and misalignment with evolving safety standards. Recognizing and addressing these pitfalls are vital to ensuring

arguments effectively contribute to the robustness and reliability of safety cases..

New column titles may be introduced in an Excel spreadsheet, headings in Word documents, or organized within collaborative platforms such as Confluence. For further details, see also Table II above: "Arguments".



Fig. 3. Arguments and related challenges

G. Synergies, similarities and conflicting requirements

When starting to comply with e.g. IEC 61508, we often recommend to first start to comply with ISO 9001 "quality management" to ensure a good foundation for the quality management. Safety and security share synergies in risk management, lifecycle integration, and resilience through defense-in-depth IEC TR 63069:2019 "Industrial-process measurement, control and automation - Framework for functional safety and security". However, conflicts arise in system isolation versus connectivity, fail-safe versus fail-secure mechanisms, and certification constraints against security updates. Addressing these challenges requires structured co-assurance strategies like the Safety-Security Assurance Framework (SSAF) [20], synchronized safety-security risk assessments, and agile validation approaches. Balancing trade-offs ensures systems remain both safe and secure while adapting to evolving risks and compliance requirements in critical industries.

ISO 42001:2023 "Information technology — Artificial intelligence — Management system" is the AI management system standard. If you already have ISO/IEC 27001 "Information security, cybersecurity and privacy protection — Information security management systems — Requirements", integrating ISO 42001 is simpler than you might expect. Both frameworks follow a management system approach, meaning half of the work is already done. Clauses related to leadership, planning, and performance evaluation are nearly identical. The same teams—Security, Privacy, Risk, Legal, Product, and Leadership—are involved in both frameworks, so no major organizational overhaul is needed. Additionally, ISO 27001 and ISO 42001 share the same audit process. This allows organizations to combine external audits and reduce compliance overhead.

There are several differences between domains, as demonstrated by Oko et al. [21] and [22]. These studies show that mapping safety integrity levels—such as SIL and ASIL (Automotive SIL) — varies significantly depending on domain-specific risk perceptions, regulatory priorities, and architectural constraints. This underscores the importance of cross-domain profiles and modular approaches, as emphasized in the CRF framework, to ensure consistent and adaptable compliance across industries.

New column titles in an excel sheet or headings in a word documents are, see also Table II below: Synergies/Conflicts.

H. Guidelines and references

Relevant guidelines for the IEC 61508 requirements series are provided in several documents. IEC 61508-5 offers examples of methods for determining safety integrity levels (SILs), including risk-based approaches and various quantitative, qualitative, and semi-quantitative techniques. IEC 61508-6 provides guidelines on applying IEC 61508-2 and IEC 61508-3, covering methods for evaluating hardware failure probabilities, diagnostic coverage, and software safety integrity. IEC 61508-7 summarizes techniques and measures for managing functional safety, focusing on the control of random hardware failures, the prevention of systematic failures, and the assurance of software safety integrity. In addition, there exists relevant standards like IEC TR 61508-6-1: draft 2025 “*Treatment of hardware and software developed to iso 26262*” and e.g. IEC TS 63394:2023 “*Safety of machinery - Guidelines on functional safety of safety-related control systems*”. The guideline “070 – *Offshore Norge Recommended Guidelines for the Application of IEC 61508 and IEC 61511 in the Norwegian Petroleum Industry*” [4] provides standardized guidance on applying the IEC 61508 and IEC 61511 standards within the Norwegian petroleum sector. It primarily focuses on defining Safety Integrity Level (SIL) requirements for safety-instrumented functions (SIFs) and offers structured methodologies for risk assessment, safety management, system validation, and functional safety assessment. The guideline aims to simplify and harmonize the use of these international standards, ensuring consistency in risk-based safety requirements.

The Data Safety Guidance [22] provides guidance on managing data safety in safety-critical systems, addressing risks associated with data rather than software or hardware. The document outlines a data safety management process based on ISO 31000:2018 “*Risk management — Guidelines*”, including principles for risk assessment, mitigation, and compliance. This latest version introduces new appendices on AI and autonomy. This document will be referenced in the next edition of IEC 61508-3 that includes software requirements.

Regarding papers and guidelines from relevant organizations we have mainly used this two documents as part of our development projects. The Carter Paper [24] discusses the challenges of using Python in flight-critical and safety-critical applications, particularly in aviation. It highlights concerns related to Python’s reliance on an interpreter, dynamic typing, and garbage collection, which introduce risks for deterministic execution and software assurance. The paper evaluates potential courses of action for mitigating these concerns, such as certifying Python, transitioning to compiled languages, or developing within a pre-certified environment. It provides guidance for machine learning (ML) practitioners developing models for deployment in aviation, aligning with

airworthiness standards like RTCA DO-178C “*Software Considerations in Airborne Systems and Equipment Certification*”. Learnings from this paper has been used when applying the w model as described by EASA [25]. Python is favored in the Machine Learning Development Lifecycle (MLDL) for its flexibility and libraries. However, its dynamic typing and interpreted nature create challenges in the Machine Learning Implementation Lifecycle (MLIL), especially for safety-critical applications. Transitioning to languages like C++ or Rust in the MLIL enhances reliability and compliance. Ongoing projects [26] explore shifting from Python in MLDL to Rust in MLIL to address these challenges.

New column titles in an excel sheet or headings in a word documents are, see also Table II below: Guidelines and references.

I. Summary of the new aspects and relevant columns in the CRF

In the table below we have summarised the relevant column titles in an excel sheet since IEC has already used excel sheet for this work, but other approaches like word or confluence can be used.

TABLE III. LINK BETWEEN CRF ASPECTS AND NEW COLUMN TITLES

Current and new aspects	Column titles
IV.A TRFs already exists for cybersecurity IEC 62443 series. See Table 1 above. This is a good starting point for all TRFs and CRFs. Part 4-1 “ <i>product development</i> ” and 4-2 “ <i>IACS components</i> ” Includes these column titles	• Conformity statement • Conformity evidence • Result – Remarks • Verdict
Part 2-4 “ <i>IACS service providers</i> ”	In addition to the above titles, these are added: • Applicant role • Declared maturity level - Applicable component
Part 3-3 “ <i>system security requirements and security levels</i> ”	In addition to the 4-1/2 titles, these are added: • Requirement name • Applicable zone
IV.B Profile approach	• Roles: This could be both internal and external roles. Example of roles are manufacturer (Original Equipment Manufacturer (OEM), Tier 1 etc) and integrator. • Domain/topics. Example of domains could be drones and topics could be tools.
IV.C Software, AI, cybersecurity and safety practices	• Practices
IV.D Linking Evidence, Work Products, and the Safety Case	• Evidence and links Named “Conformity evidence” in the current editions of the IEC 62443 TRFs
IV.E Arguments	• Arguments
IV.F Synergies, similarities and conflicting requirements	• Synergies/conflicts

Current and new aspects	Column titles
IV.G Guidelines and references	• Guidelines and references

V. CONCLUSION

This paper introduces the Compliance Report Format (CRF) as an advanced framework for managing compliance in AI-based functional safety (FuSa) and cybersecurity domains. Building upon the IEC Test Report Format (TRF), the CRF enhances traceability, consistency, and efficiency—particularly in the development of structured assurance cases, which may include safety cases as subsets. By integrating agile methodologies, profile-based tailoring, structured arguments, and explicit evidence linkages, the CRF aligns with evolving international standards such as IEC 61508, ISO/IEC TR 5469:2024, the IEC 62443 series, and UL 4600.

The CRF also supports a shift-left approach, enabling early AI risk analysis, FuSa integration, and continuous validation throughout the development lifecycle—thereby reducing compliance bottlenecks at later stages. Incorporating concepts like DoR and DoD [9], it ensures that safety and cybersecurity requirements are both well-scoped and fully validated before release. Additionally, this paper explores cross-domain synergies, regulatory challenges, and potential conflicts—offering guidance on how CRF can improve compliance efficiency, reduce documentation overhead, and support the safe deployment of AI-driven safety systems across diverse industries.

Below we revisit the research questions our five RQ and summarize our findings and suggestions:

RQ1: How can the development of a CRF reduce the burden of complying with AI, cybersecurity and Functional safety standards?

Burden is reduced due to the Profile approach. This depends on the product/system, context and project. Adapted agile practices for safety, cybersecurity and AI helps. Synergy has been seen between ISO 9001 and IEC 61508. And we have also seen and recommend that StartUps first comply to ISO 9001 requirements and later comply to safety and AI standards. There are many similarities between ISO 9001 and IEC 61508. Easy access to guidelines and relevant references has shown to be beneficial when using platforms such as Confluence. The CRF approach shift much of the compliance work from the safety case to the CRF and as such it has benefits similar to the shift left approach.

RQ2: What specific features in the CRF can enhance clarity and improve communication both internally and with certification bodies, ensuring a smoother certification process?

Arguments and early evidence and adapted DoR for discussing with the CB and other relevant stakeholders. Agile practices as listed in sub-clause IV.D above. Guidelines and references are relevant to ensure common understanding.

RQ3: How can a structured CRF effectively document and trace the rationale behind design and coding decisions to mitigate knowledge loss?

Linking safety plans, evidence, and safety cases is fundamental for maintaining traceability and consistency in compliance activities. Our review of the literature, along with

discussions held with software engineers, underscores the critical importance of constructing robust, structured arguments. This approach significantly mitigates knowledge loss, especially when engineers retire or transition to other roles. Additionally, utilizing established guidelines and well-documented references ensures a shared and clear understanding among all stakeholders involved.

RQ4: How can the use of profiles and agile practices in the CRF reduce the workload required to provide sufficient evidence while ensuring compliance with AI, cybersecurity and safety standards?

Only relevant requirements are applied.

Appoint and include in the CRF who are responsible for the different requirements early. Several adapted agile practices are relevant, see list on clause IV.D.

RQ5: How can the CRF leverage cross-domain knowledge to integrate and align requirements across different standards, such as IEC 61508 and ISO 26262, while identifying synergies and managing potential conflicts?

Analysis, guidelines and papers. In addition, there exists relevant standards like IEC TR 61508-6-1 and e.g. IEC TS 63394:2023. As part of RQ3, we have also learned from discussion with experienced engineers that have knowledge about the conflicts. This knowledge is included in the CRF template for the relevant standards.

As a next step, we plan to pilot the CRF approach in an industrial innovation project, based on IEC 61508, to quantitatively measure reductions in compliance effort and enhancements in traceability. Furthermore, we intend to engage in discussions with IECEE to explore how integrating the CRF into existing IECEE schemes could streamline international certifications and facilitate broader industry adoption.

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